1.Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Wireshark Practical: Flipkart Website

Aim: To capture network packets while refreshing Flipkart and identify the four TCP/IP layers.

Steps

1. Open Wireshark on your laptop.
2. Select your Wi-Fi network and click Start.
3. Open Flipkart and refresh the page.
4. Return to Wireshark and click Stop.
5. Check the Protocol column and packet details.

Observation Table

TCP/IP Layer	Protocol	What it does
1. Application	HTTPS / TLS	Helps load the website securely.
2. Transport	TCP	Transfers data between devices reliably.
3. Internet	IP	Helps data reach the correct destination.
4. Network Access	Ethernet / Wi-Fi	Transfers data through the local network.

2.Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

Difference Between OSI and TCP/IP Models

The OSI model has 7 layers, while the TCP/IP model has 4 layers. Both explain how data travels from one device to another over a network.

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Easy Explanation and Real-World Example
7. Application	Application	The app we use. Example: Sending a message on WhatsApp.
6. Presentation	Application	Converts data into a suitable format. Example: Sending a

		photo on WhatsApp.
5. Session	Application	Starts and maintains communication. Example: Keeping a WhatsApp call connected.
4. Transport	Transport	Delivers data to the correct app. Example: Sending a Zomato order request.
3. Network	Internet	Finds the path to the destination using IP addresses. Example: Sending an order to a Zomato server.
2. Data Link	Network Access / Link	Transfers data between devices on the local network. Example: Your phone sending data through Wi-Fi.
1. Physical	Network Access / Link	Sends data as electrical, radio, or light signals. Example: Wi-Fi signals connecting your phone to the router.

Simple conclusion

- OSI model: Has 7 layers and helps us understand how networking works.
- TCP/IP model: Has 4 layers and is used for real internet communication.
- Example: When you send a WhatsApp message, all these networking functions work together to deliver it to the other person.

**3.Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.**

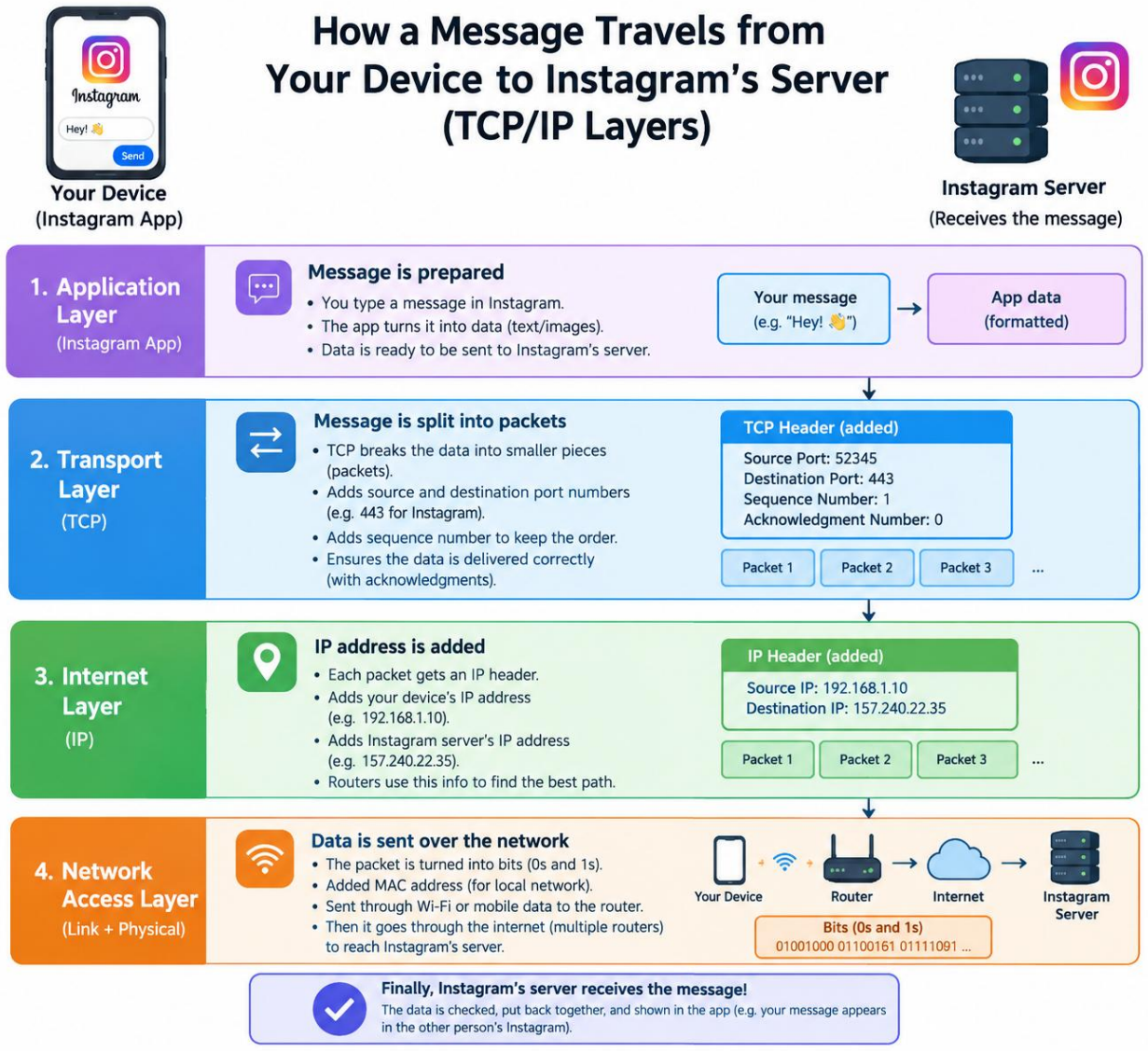
TCP Segment Header (Easy Answer)

TCP Header Field	Simple Explanation
Source Port	Shows which application is sending data.
Destination Port	Shows which application will receive data.
Sequence Number	Keeps data in the correct order.
Acknowledgment Number	Confirms the data received.
Header Length	Shows the size of the TCP header.
Flags	Help start, manage, and close a connection.
Window Size	Shows how much data the receiver can accept.
Checksum	Checks whether data has errors.
Urgent Pointer	Points to urgent data.

Options	Contains extra settings for communication.
Data (Payload)	Contains the actual message or information being sent.

Example: When you send a WhatsApp message, TCP helps send the data in the correct order and checks for errors.

4.Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5.Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Wireshark Packet Example

I captured one packet in Wireshark while using an app.

- TCP/IP Layer: Transport Layer
- How I identified it: I checked the packet details in Wireshark and found TCP, source port, and destination port.
- Real-world action: It represents sending a message on WhatsApp. TCP helps deliver the data correctly and in the right order.

Conclusion: Wireshark helps us see how data travels through different network layers.